

An Overview of Noise-Contrastive Estimation

From Local NCE to Global NCE and InfoNCE

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1 Brief Overview

General concept and motivation. In many statistical and machine learning problems, a model must estimate a probability distribution from observed data, or a conditional probability distribution relating variables. The traditional approach often relies on maximum likelihood estimation (MLE). However, for models with very large output spaces or unnormalized probability models, directly optimizing the likelihood can be impractical because it requires evaluating an expensive partition function.

Noise-contrastive estimation (NCE) addresses this difficulty by turning density estimation into a task of distinguishing real data from noise. Instead of explicitly evaluating the partition function, NCE trains a classifier to identify which samples come from the data distribution and which come from a known noise distribution. Model parameters are then learned through this classification objective.

Starting from the same contrastive principle, different posterior constructions and objectives lead to three formulations discussed here:

1. **Local NCE (binary NCE): separate binary classification decisions.** For each observed sample, the model estimates the posterior probability that it comes from the data distribution rather than noise. Evaluating individual samples substantially reduces computation. Some implementations additionally adopt an approximate self-normalization assumption; this is not a requirement of NCE in general.
2. **Global NCE (ranking NCE): classification over a candidate set.** The task becomes multiclass identification over N candidates. The candidates are compared jointly to locate the sample drawn from the data distribution. This construction naturally cancels the normalizing constant in the posterior, avoiding the self-normalization assumption used in the particular local formulation considered below.
3. **InfoNCE: optimizing a lower bound on mutual information.** InfoNCE can be viewed as a special case of Global NCE under a particular noise-sampling scheme. When the noise distribution equals the target variable’s marginal distribution, the objective is directly related to a lower bound on mutual information between the variables. This perspective provides a theoretical foundation for self-supervised and contrastive representation learning.

2 Main Discussion

2.1 Local NCE

2.1.1 The Original Problem and the Full-Softmax Bottleneck

In traditional probabilistic models, a central goal is to estimate the conditional probability of a target x given a context c . Let $s_\theta(x, c) \in \mathbb{R}$ be an unnormalized score. Define the positive quantity $f_\theta(x, c) = \exp s_\theta(x, c) > 0$. Then

$$p_\theta(x | c) = \frac{\exp s_\theta(x, c)}{\sum_{x'} \exp s_\theta(x', c)} = \frac{f_\theta(x, c)}{Z_\theta(c)}, \quad Z_\theta(c) = \sum_{x'} f_\theta(x', c). \quad (1)$$

Here:

- $s_\theta(x, c)$ is a real-valued scoring function expressing target-context compatibility;
- $f_\theta(x, c) = \exp s_\theta(x, c)$ is a positive, unnormalized mass or density;
- $Z_\theta(c)$ is the context-dependent partition function that normalizes probabilities over the full candidate space;
- θ is the model’s parameter vector.

The sums above assume a discrete target space; for continuous targets, they are replaced by integrals.

2.1.2 Reformulation as Binary Classification

To avoid evaluating $Z_\theta(c)$ over the entire candidate space, Local NCE does not directly maximize the normalized model's likelihood. Instead, density estimation is reformulated as separate binary classification decisions.

The inputs come from two sources: real data and noise. For each target x and context c , the model decides whether the pair originates from the data distribution ($D = 1$) or a known noise distribution ($D = 0$).

2.1.3 The Data-Generation Mechanism

Sampling uses two sources:

- Draw one real sample from $p_{\text{data}}(x | c)$, labeled $D = 1$.
- Independently draw k noise samples from $q(x)$, labeled $D = 0$.

With one real sample for every k noise samples, Bayes' theorem gives

$$P(D = 0 | x, c) = \frac{kq(x)}{p_{\text{data}}(x | c) + kq(x)}, \quad P(D = 1 | x, c) = \frac{p_{\text{data}}(x | c)}{p_{\text{data}}(x | c) + kq(x)}.$$

Assume $q(x) > 0$ wherever $p_{\text{data}}(x | c) > 0$, so that the relevant density ratio is defined. **Intuition:** as the data density increases relative to the noise density, the posterior probability that the sample is real increases.

2.1.4 The Normalizing Constant and Self-Normalization

The probability model has the form

$$p_\theta(x | c) = \frac{f_\theta(x, c)}{Z_\theta(c)}, \quad f_\theta(x, c) > 0. \quad (2)$$

The key idea of NCE is that the partition function need not be computed exactly by scanning the full candidate space at every optimization step. In original NCE, the log-normalizing constant can be parameterized and estimated alongside the other parameters in suitable settings. For conditional models, $Z_\theta(c)$ may depend on each context, so original NCE should not be equated with the assumption $Z_\theta(c) = 1$.

Some implementations introduce a self-normalization assumption or regularization so that

$$Z_\theta(c) \approx 1.$$

Under this assumption, $p_\theta(x | c) \approx f_\theta(x, c)$. The model posteriors used to approximate the classification problem are then

$$\begin{aligned} \text{(a)} \quad P_\theta(D = 0 | x, c) &\approx \frac{kq(x)}{f_\theta(x, c) + kq(x)}, \\ \text{(b)} \quad P_\theta(D = 1 | x, c) &\approx \frac{f_\theta(x, c)}{f_\theta(x, c) + kq(x)}. \end{aligned}$$

These are model posteriors. The ideal Bayes posteriors in the preceding section depend on the true data distribution $p_{\text{data}}(x | c)$.

2.1.5 Deriving (a) and (b) from Bayes' Theorem

Bayes' theorem states

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}. \quad (3)$$

Choosing a sample uniformly from the mixture of one real sample and k noise samples gives

$$P(D = 1) = \frac{1}{k+1}, \quad P(D = 0) = \frac{k}{k+1}. \quad (4)$$

The source-conditional densities are

$$p(x | D = 1, c) = p_{\text{data}}(x | c), \quad p(x | D = 0, c) = q(x). \quad (5)$$

For a sample x in context c ,

$$P(D = 1 | x, c) = \frac{p(x | D = 1, c)P(D = 1)}{p(x | c)}. \quad (6)$$

Expanding the denominator by the law of total probability,

$$p(x | c) = p(x | D = 1, c)P(D = 1) + p(x | D = 0, c)P(D = 0), \quad (7)$$

$$= p_{\text{data}}(x | c) \frac{1}{k+1} + q(x) \frac{k}{k+1} = \frac{p_{\text{data}}(x | c) + kq(x)}{k+1}. \quad (8)$$

Substituting into Bayes' theorem,

$$P(D = 1 | x, c) = \frac{p_{\text{data}}(x | c)}{p_{\text{data}}(x | c) + kq(x)}, \quad (9)$$

$$P(D = 0 | x, c) = 1 - P(D = 1 | x, c) = \frac{kq(x)}{p_{\text{data}}(x | c) + kq(x)}. \quad (10)$$

Equations (9)–(10) are the ideal Bayes posteriors for the sampling mechanism. During training, $p_{\text{data}}(x | c)$ is not explicitly known, so NCE substitutes a model $p_{\theta}(x | c)$. If self-normalization $Z_{\theta}(c) \approx 1$ is also assumed, then $p_{\theta}(x | c) \approx f_{\theta}(x, c)$, yielding model posteriors (a) and (b).

2.1.6 The Objective Function

The objective maximizes the classification log-likelihood, increasing the probability of correctly identifying both real and noise samples:

- Real sample: $\log P_{\theta}(D = 1 | x, c)$.
- Noise samples: $\log P_{\theta}(D = 0 | x'_i, c)$.

For one data pair, the sampled objective is

$$\log P_{\theta}(D = 1 | x, c) + \sum_{i=1}^k \log P_{\theta}(D = 0 | x'_i, c), \quad x'_i \stackrel{\text{iid}}{\sim} q. \quad (11)$$

The noise term can be rewritten as

$$\sum_{i=1}^k \log P_{\theta}(D = 0 | x'_i, c) = k \left[\frac{1}{k} \sum_{i=1}^k \log P_{\theta}(D = 0 | x'_i, c) \right]. \quad (12)$$

For fixed θ, c, k , its expectation is exactly

$$\mathbb{E}_{x'_{1:k} \sim q^k} \left[\sum_{i=1}^k \log P_{\theta}(D = 0 | x'_i, c) \right] = k \mathbb{E}_{x' \sim q} [\log P_{\theta}(D = 0 | x', c)]. \quad (13)$$

Thus the expected objective is

$$J_{\text{localNCE}} = \sum_{(x,c) \in \mathcal{D}} [\log P_{\theta}(D = 1 | x, c) + k \mathbb{E}_{x' \sim q} \log P_{\theta}(D = 0 | x', c)]. \quad (14)$$

Here $(x, c) \in \mathcal{D}$ is a real training pair, $x' \sim q$ is noise, and k is the configured noise-to-data ratio. Exact evaluation of the expectation can itself require a sum or integral over the full candidate space. It is therefore usually estimated by Monte Carlo:

$$J_{\text{localNCE,MC}} = \sum_{(x,c) \in \mathcal{D}} \left[\log P_{\theta}(D = 1 \mid x, c) + \sum_{i=1}^k \log P_{\theta}(D = 0 \mid x'_i, c) \right]. \quad (15)$$

This estimator is unbiased. With k fixed in the posterior, an average over an increasing number of independent noise draws converges to the corresponding expectation under the usual integrability conditions. The law of large numbers applies to that average, not directly to an unnormalized sum.

2.1.7 Adaptive Gradient Weighting

For a real sample x in context c and k noise samples x_i , the gradient of the sampled objective is

$$\begin{aligned} \nabla_{\theta} J_{c,x}(\theta) &= \frac{kq(x)}{p_{\theta}(x \mid c) + kq(x)} \nabla_{\theta} \log p_{\theta}(x \mid c) \\ &\quad - \sum_{i=1}^k \frac{p_{\theta}(x_i \mid c)}{p_{\theta}(x_i \mid c) + kq(x_i)} \nabla_{\theta} \log p_{\theta}(x_i \mid c). \end{aligned} \quad (16)$$

The notation is:

- θ : parameters to optimize; c : input context;
- x : real target (positive sample); x_i : the i th noise sample drawn from q ;
- k : number of noise samples per real sample;
- $J_{c,x}(\theta)$: the local objective for one sampling step;
- $\nabla_{\theta} J_{c,x}(\theta)$: its parameter gradient;
- $p_{\theta}(x \mid c)$: the model probability or density assigned to x in context c ;
- $q(x)$ and $q(x_i)$: the noise distribution evaluated at the real and noise samples;
- $\nabla_{\theta} \log p_{\theta}(x \mid c)$ and $\nabla_{\theta} \log p_{\theta}(x_i \mid c)$: the corresponding base gradients.

The two fractions are adaptive weighting coefficients. They scale each base gradient according to how well the model distinguishes data from noise. Equation (16) assumes that q is fixed with respect to θ . If the implemented classifier uses f_{θ} directly, the same formula holds with p_{θ} replaced throughout by f_{θ} .

A. Positive-gradient coefficient:

$$\text{Positive coefficient} = \frac{kq(x)}{p_{\theta}(x \mid c) + kq(x)}. \quad (17)$$

If the model assigns a very low value to the real sample, $p_{\theta}(x \mid c) \ll kq(x)$, the coefficient approaches 1 and preserves most of its base gradient. If the model already assigns a high value relative to $kq(x)$, the coefficient approaches 0. This term is evaluated at the real target rather than summed over the full vocabulary.

B. Negative-gradient coefficient:

$$\text{Negative coefficient} = \frac{p_{\theta}(x_i \mid c)}{p_{\theta}(x_i \mid c) + kq(x_i)}. \quad (18)$$

This coefficient controls the correction for each noise sample. If $p_{\theta}(x_i \mid c) \gg kq(x_i)$, the coefficient approaches 1, preserving the corrective base gradient. If the noise sample is already identified correctly with high confidence, the coefficient approaches 0. Optimization therefore assigns larger coefficients to harder noise samples.

Remark. These coefficients provide adaptive weighting: classification errors receive larger weights, while confidently correct examples receive smaller ones. This does not by itself guarantee larger parameter-gradient norms, which also depend on the base gradients. Avoiding the full partition function requires an

NCE parameterization that learns or approximates normalization; inserting an exactly normalized softmax into (16) would retain its computational cost.

Notation convention. In this section, c denotes context, x denotes the target, and q denotes the noise distribution. In the following sections, x denotes context and y denotes the target; the underlying relationships are unchanged.

2.2 Global NCE

2.2.1 The Original Problem and Motivation

Where Local NCE uses separate binary decisions, Global NCE reformulates the task as ranking or multiclass classification over a candidate set.

Rather than evaluating every sample independently, Global NCE compares candidates in a shared context. Given exactly one real sample and $N - 1$ noise samples, the model identifies the position containing the real sample.

2.2.2 The Joint Data-Generation Mechanism

Global NCE sampling defines a joint distribution over the index, the real pair, and the noise candidates:

1. Draw an index $i \in \{1, \dots, N\}$ uniformly, so $P(i) = 1/N$.
2. At position i , draw a real pair $(x, y_i) \sim p_{XY}^{\text{op}}$, where x is context and y_i is the target.
3. At every position $j \neq i$, independently draw $y_j \sim q_Y$.

The joint distribution is

$$p_{IXY^N}^{\text{op}, q_Y}(i, x, y_1, \dots, y_N) := \frac{1}{N} p_{XY}^{\text{op}}(x, y_i) \prod_{j \neq i} q_Y(y_j). \quad (19)$$

Here $1/N$ is the prior probability of the real sample's position; $p_{XY}^{\text{op}}(x, y_i) = p_X^{\text{op}}(x) p_{Y|X}^{\text{op}}(y_i | x)$ is the real-pair density; and $\prod_{j \neq i} q_Y(y_j)$ is the joint density of the independent noise samples. The superscript op denotes the true population distribution.

2.2.3 The Bayes Posterior over Candidate Indices

After observing context x and all N candidates, the posterior probability that position i contains the real sample is

$$p_{I|XY^N}^{\text{op}, q_Y}(i | x, y_{1:N}) = \frac{p_{Y|X}^{\text{op}}(y_i | x) \prod_{j \neq i} q_Y(y_j)}{\sum_{\ell=1}^N p_{Y|X}^{\text{op}}(y_\ell | x) \prod_{j \neq \ell} q_Y(y_j)} = \frac{\frac{p_{Y|X}^{\text{op}}(y_i | x)}{q_Y(y_i)}}{\sum_{\ell=1}^N \frac{p_{Y|X}^{\text{op}}(y_\ell | x)}{q_Y(y_\ell)}}. \quad (20)$$

The density ratio $p_{Y|X}^{\text{op}}(y_i | x)/q_Y(y_i)$ measures how compatible a candidate is with the conditional data distribution relative to the background noise distribution. Assume $q_Y(y) > 0$ on the support of $p_{Y|X}^{\text{op}}(\cdot | x)$.

Algebraic simplification. Divide numerator and denominator by $\prod_{m=1}^N q_Y(y_m)$. The cross-products of noise densities cancel, leaving normalized density ratios, equivalently a softmax over their logarithms.

2.2.4 The Objective Function

Global NCE uses an unnormalized score $s_\theta(x, y)$ and a softmax over the candidate set:

$$p_{I|XY^N}^\theta(i | x, y_{1:N}) := \frac{\exp s_\theta(x, y_i)}{\sum_{j=1}^N \exp s_\theta(x, y_j)}, \quad i \in \{1, \dots, N\}. \quad (21)$$

Training minimizes cross-entropy, equivalently maximizing the log-likelihood of the correct index:

$$\mathcal{L}_{\text{GlobalNCE}}(\theta) = \mathbb{E}_{(i, x, y_{1:N}) \sim p_{IXY^N}^{\text{op}, q_Y}} \left[-\log \frac{\exp s_\theta(x, y_i)}{\sum_{j=1}^N \exp s_\theta(x, y_j)} \right]. \quad (22)$$

2.2.5 Advantages over Local NCE

For a sufficiently expressive model class, cross-entropy is minimized when the predicted index distribution matches the Bayes posterior $p(I | x, y_{1:N})$. This does not directly mean that $p_\theta(y | x) = p_{\text{data}}(y | x)$: classification learns a density ratio. At an ideal optimum, the score has the form

$$s_\theta^{q_Y}(x, y) = \log \frac{p_{Y|X}^{\text{op}}(y | x)}{q_Y(y)} + \log C_x, \quad C_x > 0. \quad (\text{BT1})$$

This characterization holds on the relevant support, with limiting scores where the density ratio is zero. Since softmax is invariant to a shared additive constant, the context-only term cancels. Thus Global NCE does not require the self-normalization approximation $Z(c) \approx 1$ used in the local implementation considered earlier. Binary NCE itself need not impose that approximation.

2.2.6 Cancellation of the Normalizing Constant

For a general softmax,

$$p(i) = \frac{\exp(\text{logit}_i)}{\sum_{j=1}^N \exp(\text{logit}_j)}. \quad (23)$$

Adding the same constant a to all logits gives

$$p(i) = \frac{\exp(\text{logit}_i + a)}{\sum_{j=1}^N \exp(\text{logit}_j + a)} = \frac{\exp(\text{logit}_i) \exp(a)}{\exp(a) \sum_{j=1}^N \exp(\text{logit}_j)}, \quad (24)$$

$$= \frac{\exp(\text{logit}_i)}{\sum_{j=1}^N \exp(\text{logit}_j)}. \quad (25)$$

Write the true conditional data distribution in terms of a score and partition function:

$$p_{Y|X}^{\text{op}}(y | x) = \frac{\exp S^{\text{true}}(x, y)}{Z(x)}. \quad (\text{BT2})$$

Substituting (BT2) into (BT1),

$$S^\theta(x, y) = \log \left(\frac{\exp S^{\text{true}}(x, y)}{Z(x) q_Y(y)} \right) + \log C_x, \quad (26)$$

$$S^\theta(x, y) = S^{\text{true}}(x, y) - \log q_Y(y) + [\log C_x - \log Z(x)]. \quad (27)$$

Here $S^{\text{true}}(x, y)$ is the true-data compatibility score, while the bracketed term depends only on context x .

For a fixed x , this term is added equally to every candidate's logit. It cancels when softmax is applied across candidates. Consequently,

$$S^\theta(x, y) \equiv S^{\text{true}}(x, y) - \log q_Y(y) \quad \text{up to a context-only additive term.} \quad (28)$$

2.3 InfoNCE

In Global NCE, the model classifies a set of N candidates: one real sample y_i and $N - 1$ noise samples y_j . The noise distribution q_Y is a design choice, such as a uniform or unigram distribution. The standard InfoNCE construction considered here is the special case

$$q_Y(y) \equiv p_Y^{\text{op}}(y). \quad (29)$$

The model distinguishes a dependent pair $(x, y) \sim p_{XY}^{\text{op}}$ from targets drawn independently from the marginal. This construction connects the optimal score directly to the pair's pointwise mutual information (PMI).

2.3.1 The Optimal Score When Noise Equals the Marginal

From (BT1), setting $q_Y(y) = p_Y^{\text{op}}(y)$ yields

$$s_{\theta}^*(x, y) = \log \left[\frac{p_{Y|X}^{\text{op}}(y | x)}{p_Y^{\text{op}}(y)} \right] + \log C_x. \quad (30)$$

By the definition of conditional probability,

$$p_{Y|X}^{\text{op}}(y | x) = \frac{p_{XY}^{\text{op}}(x, y)}{p_X^{\text{op}}(x)}. \quad (31)$$

Therefore,

$$s_{\theta}^*(x, y) = \log \left[\frac{p_{XY}^{\text{op}}(x, y)}{p_X^{\text{op}}(x)p_Y^{\text{op}}(y)} \right] + \log C_x. \quad (32)$$

The logarithmic term is PMI:

$$\text{PMI}(x, y) = \log \left[\frac{p_{XY}^{\text{op}}(x, y)}{p_X^{\text{op}}(x)p_Y^{\text{op}}(y)} \right]. \quad (33)$$

When the critic can represent the Bayes solution, its optimal score equals PMI plus a context-only constant. That constant cancels in softmax and does not affect candidate ranking.

- If X and Y are statistically independent, $p(x, y) = p(x)p(y)$ and $\text{PMI}(x, y) = 0$. After removing the common constant, PMI gives no preference among candidates.
- If $p(x, y) > p(x)p(y)$, then $\text{PMI}(x, y) > 0$. The score is above the independence baseline, reflecting greater-than-expected co-occurrence.
- If $p(x, y) < p(x)p(y)$, then $\text{PMI}(x, y) < 0$. The score is below the independence baseline, reflecting less-than-expected co-occurrence.

2.4 InfoNCE as a Lower Bound on Mutual Information

2.4.1 Proof Using an Index Variable and KL Divergence

Keep the Global NCE sampling mechanism, but set $q_Y(y) = p_Y(y)$. Introduce the index variable

$$I \sim \text{Uniform}\{1, \dots, N\}.$$

For each sampling step:

- Draw $X \sim p_X$.
- At position I , draw the positive $Y_I \sim p_{Y|X}(\cdot | X)$.
- For every $j \neq I$, independently draw $Y_j \sim p_Y$.

The resulting joint distribution is

$$P(i, x, y_{1:N}) = \frac{1}{N} p_X(x) p_{Y|X}(y_i | x) \prod_{j \neq i} p_Y(y_j). \quad (34)$$

Define a reference distribution in which all candidates are independently drawn from the marginal:

$$Q(i, x, y_{1:N}) = \frac{1}{N} p_X(x) \prod_{j=1}^N p_Y(y_j). \quad (35)$$

Their density ratio is

$$\frac{P(i, x, y_{1:N})}{Q(i, x, y_{1:N})} = \frac{p_{Y|X}(y_i | x)}{p_Y(y_i)}. \quad (36)$$

Since $(X, Y_I) \sim p_{XY}$ under P , their KL divergence equals mutual information:

$$\begin{aligned} D_{\text{KL}}(P \| Q) &= \mathbb{E}_P \left[\log \frac{P(I, X, Y_{1:N})}{Q(I, X, Y_{1:N})} \right] \\ &= \mathbb{E}_{p_{XY}} \left[\log \frac{p_{Y|X}(Y | X)}{p_Y(Y)} \right] = I(X; Y). \end{aligned} \quad (37)$$

We assume the relevant density ratios and expectations are well defined; all logarithms are natural, so information is measured in nats.

Step 1: Apply the KL chain rule. Let $Z = (X, Y_{1:N})$. Decomposing over Z and I gives

$$D_{\text{KL}}(P_{I,Z} \| Q_{I,Z}) = D_{\text{KL}}(P_Z \| Q_Z) + \mathbb{E}_{Z \sim P_Z} D_{\text{KL}}(P(I | Z) \| Q(I | Z)). \quad (38)$$

By nonnegativity of KL,

$$I(X; Y) \geq \mathbb{E}_{Z \sim P_Z} D_{\text{KL}}(P(I | Z) \| Q(I | Z)). \quad (39)$$

Step 2: Find the index posterior under the reference distribution. Under Q , every Y_j comes from the same marginal and is independent of I . Thus

$$Q(I = i | Z) = \frac{1}{N}. \quad (40)$$

For a fixed observation $Z = z$,

$$D_{\text{KL}}(P(I | Z = z) \| \text{Uniform}[N]) = \sum_{i=1}^N P(i | z) \log \frac{P(i | z)}{1/N} = \log N - H_P(I | Z = z). \quad (41)$$

Taking the expectation over Z and substituting into (39),

$$I(X; Y) \geq \log N - H_P(I | X, Y_{1:N}). \quad (42)$$

Step 3: Relate this to cross-entropy and InfoNCE. Let the critic-induced posterior be

$$p_\theta(i | x, y_{1:N}) = \frac{\exp s_\theta(x, y_i)}{\sum_{j=1}^N \exp s_\theta(x, y_j)}. \quad (43)$$

The InfoNCE loss is the expected cross-entropy for predicting the true index:

$$\mathcal{L}_{\text{InfoNCE}}(\theta) = \mathbb{E}_P[-\log p_\theta(I | X, Y_{1:N})]. \quad (44)$$

Cross-entropy decomposes into conditional entropy plus KL:

$$\begin{aligned} \mathcal{L}_{\text{InfoNCE}}(\theta) &= H_P(I | X, Y_{1:N}) \\ &\quad + \mathbb{E}_{Z \sim P_Z} D_{\text{KL}}(P(I | Z) \| p_\theta(I | Z)) \\ &\geq H_P(I | X, Y_{1:N}). \end{aligned} \quad (45)$$

Combining (42) and (45) yields the InfoNCE lower bound:

$$\boxed{I(X;Y) \geq \log N - \mathcal{L}_{\text{InfoNCE}}(\theta)}. \quad (46)$$

If the critic matches the Bayes posterior, the final KL term in (45) vanishes and $\mathcal{L}_{\text{InfoNCE}} = H_P(I \mid X, Y_{1:N})$. An imperfect critic has larger cross-entropy and hence a looser bound, but (46) still holds. Even a Bayes-optimal critic need not make (46) tight, because $D_{\text{KL}}(P_Z \parallel Q_Z)$ in (38) can remain positive.

Meaning of $\log N$. Since $\mathcal{L}_{\text{InfoNCE}} \geq 0$, the lower bound cannot exceed $\log N$. This does not mean that true mutual information, or the information a representation can contain, is limited by $\log N$. If $I(X;Y) > \log N$, InfoNCE with N candidates cannot certify its full value through this bound. Increasing N raises the bound's ceiling.